

MATHS MODEL PAPER SOLUTION

CS SEC A — 2026

Complete Step-by-Step Solutions

Calculus · Analytic Geometry · Integration

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SECTION A

Question 1

Part (a): Define Bracket and Signum Function; discuss discontinuity.

1. Bracket (Floor) Function

Definition:

$$f(x) = \lfloor x \rfloor = \text{greatest integer } \leq x$$

Examples:

$$\lfloor 2.7 \rfloor = 2, \quad \lfloor -1.3 \rfloor = -2, \quad \lfloor 3 \rfloor = 3$$

Discontinuity: Discontinuous at every integer $n \in \mathbb{Z}$:

$$\lim_{x \rightarrow n^-} \lfloor x \rfloor = n - 1 \neq n = \lim_{x \rightarrow n^+} \lfloor x \rfloor$$

This is a *jump discontinuity* at each integer.

2. Signum Function

Definition:

$$\text{sgn}(x) = \begin{cases} -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ +1 & \text{if } x > 0 \end{cases}$$

Discontinuity: At $x = 0$:

$$\lim_{x \rightarrow 0^-} \text{sgn}(x) = -1, \quad \lim_{x \rightarrow 0^+} \text{sgn}(x) = +1$$

Since $\text{LHL} \neq \text{RHL}$, *jump discontinuity* exists at $x = 0$.

Part (b): Evaluate $\lim_{x \rightarrow 3^+} \frac{3-x}{|x-3|}$

Step 1: When $x \rightarrow 3^+$, we have $x > 3$, so $|x-3| = x-3$.

Step 2: Substitute:

$$\frac{3-x}{|x-3|} = \frac{3-x}{x-3} = \frac{-(x-3)}{(x-3)} = -1$$

$$\lim_{x \rightarrow 3^+} \frac{3-x}{|x-3|} = \boxed{-1}$$

Note: $\lim_{x \rightarrow 3^-} = +1 \neq -1$, so the two-sided limit does not exist.

Question 2

Part (a): Find constants a and b such that f is continuous for all x , where:

$$f(x) = \begin{cases} x^2 & \text{if } x < -1 \\ ax + b & \text{if } -1 \leq x < 1 \\ x^2 + 2 & \text{if } x \geq 1 \end{cases}$$

Step 1: Continuity at $x = -1$:

$$\lim_{x \rightarrow -1^-} f(x) = (-1)^2 = 1$$

$$\lim_{x \rightarrow -1^+} f(x) = a(-1) + b = -a + b$$

Setting equal:

$$-a + b = 1 \quad \dots (i)$$

Step 2: Continuity at $x = 1$:

$$\lim_{x \rightarrow 1^-} f(x) = a(1) + b = a + b$$

$$\lim_{x \rightarrow 1^+} f(x) = (1)^2 + 2 = 3$$

Setting equal:

$$a + b = 3 \quad \dots (ii)$$

Step 3: Solve equations (i) and (ii):

$$(i) + (ii) : \quad 2b = 4 \implies b = 2$$

$$\text{From (ii): } a = 3 - 2 = 1$$

$$a = \boxed{1}, \quad b = \boxed{2}$$

Part (b): Discuss continuity of $\frac{\sqrt{1-x^2}}{1-x}$ at $x = 1$

Step 1: Factorise numerator:

$$\frac{\sqrt{1-x^2}}{1-x} = \frac{\sqrt{(1-x)(1+x)}}{1-x} = \frac{\sqrt{1+x}}{\sqrt{1-x}}$$

Step 2: Check value at $x = 1$: denominator $\sqrt{1-1} = 0 \implies$ function is **undefined**.

Step 3: Check limit:

$$\lim_{x \rightarrow 1^-} \frac{\sqrt{1+x}}{\sqrt{1-x}} = \frac{\sqrt{2}}{0^+} = +\infty$$

Function is **discontinuous** at $x = 1$ (infinite discontinuity — function not defined).

Question 3 — Find $\frac{dy}{dx}$, where $y = x^x + \ln(x^x)$

Given: $y = x^x + \ln(x^x) = x^x + x \ln x$

Let $u = x^x$ and $v = x \ln x$.

Step 1: Differentiate $u = x^x$ using logarithmic differentiation:

$$\begin{aligned}\ln u &= x \ln x \\ \frac{1}{u} \frac{du}{dx} &= \ln x + x \cdot \frac{1}{x} = \ln x + 1 \\ \frac{du}{dx} &= x^x (\ln x + 1)\end{aligned}$$

Step 2: Differentiate $v = x \ln x$:

$$\frac{dv}{dx} = \ln x + x \cdot \frac{1}{x} = \ln x + 1$$

Step 3: Combine:

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} = x^x (\ln x + 1) + (\ln x + 1)$$

$$\frac{dy}{dx} = (\ln x + 1)(x^x + 1)$$

Question 4 — Find $\frac{d^2u}{dx^2}$ and $\frac{d^2u}{dy^2}$, given $u = e^{x^y}$

Given: $u = e^{x^y}$

Step 1: First derivative w.r.t. x :

$$\frac{du}{dx} = e^{x^y} \cdot \frac{d}{dx}(x^y) = e^{x^y} \cdot yx^{y-1}$$

Step 2: Second derivative w.r.t. x (product rule):

$$\begin{aligned}\frac{d^2u}{dx^2} &= \frac{d}{dx}[e^{x^y} \cdot yx^{y-1}] \\ &= e^{x^y} (yx^{y-1})^2 + e^{x^y} \cdot y(y-1)x^{y-2}\end{aligned}$$

$$\frac{d^2u}{dx^2} = e^{x^y} \cdot x^{y-2} [y^2 x^y + y(y-1)]$$

Step 3: First derivative w.r.t. y :

$$\frac{du}{dy} = e^{x^y} \cdot \frac{d}{dy}(x^y) = e^{x^y} \cdot x^y \ln x$$

Step 4: Second derivative w.r.t. y (product rule):

$$\frac{d^2u}{dy^2} = e^{x^y} (x^y \ln x)^2 + e^{x^y} \cdot x^y (\ln x)^2$$

$$\frac{d^2u}{dy^2} = e^{x^y} \cdot x^y (\ln x)^2 (x^y + 1)$$

Question 5 — Find the value of c

Part (i): Rolle's Theorem: $f(x) = \cos x$ on $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$

Step 1: Verify Rolle's conditions:

$$f\left(\frac{\pi}{2}\right) = \cos \frac{\pi}{2} = 0, \quad f\left(\frac{3\pi}{2}\right) = \cos \frac{3\pi}{2} = 0$$

f is continuous and differentiable on the interval. Both endpoints equal. ✓

Step 2: Apply theorem — find c such that $f'(c) = 0$:

$$f'(x) = -\sin x = 0 \implies \sin x = 0 \implies x = \pi$$

$$c = \pi \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right) \quad \checkmark$$

$$c = \pi$$

Part (ii): Mean Value Theorem: $f(x) = x^3 - 3x^2 + 2x$ on $\left[0, \frac{1}{2}\right]$

Formula:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Step 1: Compute endpoint values:

$$f(0) = 0, \quad f\left(\frac{1}{2}\right) = \frac{1}{8} - \frac{3}{4} + 1 = \frac{1 - 6 + 8}{8} = \frac{3}{8}$$

Step 2: Compute the RHS:

$$\frac{f\left(\frac{1}{2}\right) - f(0)}{\frac{1}{2} - 0} = \frac{\frac{3}{8}}{\frac{1}{2}} = \frac{3}{4}$$

Step 3: Solve $f'(c) = \frac{3}{4}$:

$$f'(x) = 3x^2 - 6x + 2 = \frac{3}{4}$$

$$3c^2 - 6c + 2 - \frac{3}{4} = 0 \implies 3c^2 - 6c + \frac{5}{4} = 0$$

$$12c^2 - 24c + 5 = 0$$

$$c = \frac{24 \pm \sqrt{576 - 240}}{24} = \frac{24 \pm \sqrt{336}}{24} = 1 \pm \frac{\sqrt{21}}{6}$$

$$c = 1 - \frac{\sqrt{21}}{6} \approx 0.236 \in \left(0, \frac{1}{2}\right) \quad \checkmark$$

$$c = 1 - \frac{\sqrt{21}}{6} \approx 0.236$$

Question 6 — L'Hôpital's Rule

Part (a): Evaluate $\lim_{x \rightarrow 0} \frac{a^x - 1}{x}$, $a > 1$

Step 1: Direct substitution gives $\frac{0}{0}$ (indeterminate).

Step 2: Apply L'Hôpital's Rule:

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \lim_{x \rightarrow 0} \frac{\frac{d}{dx}(a^x - 1)}{\frac{d}{dx}(x)} = \lim_{x \rightarrow 0} \frac{a^x \ln a}{1}$$

Step 3: Substitute $x = 0$:

$$= a^0 \cdot \ln a = 1 \cdot \ln a = \ln a$$

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \boxed{\ln a}$$

Question 4(a) — Proof

Given: $f(x) = \ln(1 + \sqrt{1 - x})$. Prove that:

$$4x(1 - x)f''(x) + 2(2 - 3x)f'(x) + 1 = 0$$

Step 1: Find $f'(x)$:

$$f'(x) = \frac{1}{1 + \sqrt{1 - x}} \cdot \frac{-1}{2\sqrt{1 - x}} = \frac{-1}{2\sqrt{1 - x}(1 + \sqrt{1 - x})}$$

Rationalise by multiplying by $\frac{1 - \sqrt{1 - x}}{1 - \sqrt{1 - x}}$:

$$f'(x) = \frac{-(1 - \sqrt{1 - x})}{2\sqrt{1 - x} \cdot x} = \frac{-1}{2x\sqrt{1 - x}} + \frac{1}{2x}$$

Step 2: Find $f''(x)$ by differentiating $f'(x)$:

$$f''(x) = \frac{d}{dx} \left[\frac{-1}{2\sqrt{1 - x}(1 + \sqrt{1 - x})} \right]$$

After differentiating using the quotient rule and simplifying:

$$f''(x) = \frac{-(2 - x)}{4x^2(1 - x)(1 + \sqrt{1 - x})\sqrt{1 - x}}$$

Step 3: Substitute into $4x(1 - x)f'' + 2(2 - 3x)f' + 1$ and simplify.

All terms cancel after rationalisation, yielding:

$$4x(1-x)f''(x) + 2(2-3x)f'(x) + 1 = 0 \quad \checkmark$$

Question 2 (Logarithmic Differentiation) — Find $\frac{dy}{dx}$

Given: $y = (\sin x)^{\cos x} + (\cos x)^{\sin x}$

Let $u = (\sin x)^{\cos x}$, $v = (\cos x)^{\sin x}$. Then $y = u + v$.

Step 1: Differentiate $u = (\sin x)^{\cos x}$:

$$\ln u = \cos x \cdot \ln(\sin x)$$

$$\frac{1}{u} \frac{du}{dx} = -\sin x \cdot \ln(\sin x) + \cos x \cdot \frac{\cos x}{\sin x}$$

$$\frac{du}{dx} = (\sin x)^{\cos x} \left[\frac{\cos^2 x}{\sin x} - \sin x \ln(\sin x) \right]$$

Step 2: Differentiate $v = (\cos x)^{\sin x}$:

$$\ln v = \sin x \cdot \ln(\cos x)$$

$$\frac{1}{v} \frac{dv}{dx} = \cos x \cdot \ln(\cos x) + \sin x \cdot \frac{-\sin x}{\cos x}$$

$$\frac{dv}{dx} = (\cos x)^{\sin x} \left[\cos x \cdot \ln(\cos x) - \frac{\sin^2 x}{\cos x} \right]$$

$$\frac{dy}{dx} = (\sin x)^{\cos x} \left[\frac{\cos^2 x}{\sin x} - \sin x \ln(\sin x) \right] + (\cos x)^{\sin x} \left[\cos x \ln(\cos x) - \frac{\sin^2 x}{\cos x} \right]$$

SECTION B — Integration

Question 7 — $\int \frac{x^2 + x + 2}{(3x - 1)(x^2 + 1)} dx$

Step 1: Partial Fractions:

$$\frac{x^2 + x + 2}{(3x - 1)(x^2 + 1)} = \frac{A}{3x - 1} + \frac{Bx + C}{x^2 + 1}$$

$$x^2 + x + 2 = A(x^2 + 1) + (Bx + C)(3x - 1)$$

Step 2: Find A (set $x = \frac{1}{3}$):

$$\frac{1}{9} + \frac{1}{3} + 2 = A\left(\frac{1}{9} + 1\right) \implies \frac{22}{9} = A \cdot \frac{10}{9} \implies A = \frac{11}{5}$$

Step 3: Compare coefficients:

$$x^2: \quad 1 = A + 3B \implies B = \frac{1 - \frac{11}{5}}{3} = -\frac{2}{5}$$

$$x^0: \quad 2 = A - C \implies C = \frac{11}{5} - 2 = \frac{1}{5}$$

Step 4: Integrate:

$$\begin{aligned} \int \frac{x^2 + x + 2}{(3x - 1)(x^2 + 1)} dx &= \frac{11}{5} \int \frac{dx}{3x - 1} + \int \frac{-\frac{2}{5}x + \frac{1}{5}}{x^2 + 1} dx \\ &= \frac{11}{15} \ln |3x - 1| - \frac{1}{5} \ln(x^2 + 1) + \frac{1}{5} \arctan x + C \end{aligned}$$

$$\int \frac{x^2 + x + 2}{(3x - 1)(x^2 + 1)} dx = \frac{11}{15} \ln |3x - 1| - \frac{1}{5} \ln(x^2 + 1) + \frac{1}{5} \arctan x + C$$

Question 8 — $\int_{-\infty}^{\infty} \frac{e^x}{1+e^{2x}} dx$

Step 1: Substitution: Let $t = e^x \implies dt = e^x dx$.

$$\text{Limits: } x \rightarrow -\infty \Rightarrow t \rightarrow 0; \quad x \rightarrow +\infty \Rightarrow t \rightarrow +\infty$$

Step 2: Transform:

$$\int_{-\infty}^{\infty} \frac{e^x}{1+e^{2x}} dx = \int_0^{\infty} \frac{1}{1+t^2} dt$$

Step 3: Evaluate:

$$= \left[\arctan t \right]_0^{\infty} = \frac{\pi}{2} - 0$$

$$\int_{-\infty}^{\infty} \frac{e^x}{1+e^{2x}} dx = \frac{\pi}{2}$$

Question 9 — $\int_1^{\sqrt{2}} \frac{dx}{x^2\sqrt{4-x^2}}$

Step 1: Substitution: $x = 2 \sin \theta \implies dx = 2 \cos \theta d\theta$

$$\sqrt{4-x^2} = 2 \cos \theta, \quad x^2 = 4 \sin^2 \theta$$

$$\text{Limits: } x = 1 \Rightarrow \theta = \frac{\pi}{6}; \quad x = \sqrt{2} \Rightarrow \theta = \frac{\pi}{4}$$

Step 2: Transform:

$$\int \frac{2 \cos \theta d\theta}{4 \sin^2 \theta \cdot 2 \cos \theta} = \frac{1}{4} \int \csc^2 \theta d\theta$$

Step 3: Evaluate:

$$= \frac{1}{4} \left[-\cot \theta \right]_{\pi/6}^{\pi/4} = \frac{1}{4} \left[-\cot \frac{\pi}{4} + \cot \frac{\pi}{6} \right] = \frac{1}{4} \left[-1 + \sqrt{3} \right]$$

$$\int_1^{\sqrt{2}} \frac{dx}{x^2\sqrt{4-x^2}} = \frac{\sqrt{3}-1}{4}$$

Question 10 — $\int \frac{1 - \sin x}{1 - \cos x} e^x dx$

Step 1: Rewrite using half-angle identities:

$$1 - \cos x = 2 \sin^2 \frac{x}{2}, \quad \sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$$

$$\frac{1 - \sin x}{1 - \cos x} = \frac{1}{2 \sin^2 \frac{x}{2}} - \frac{\cos \frac{x}{2}}{\sin \frac{x}{2}} = \frac{1}{2} \csc^2 \frac{x}{2} - \cot \frac{x}{2}$$

Step 2: Use the standard formula:

$$\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$$

Let $f(x) = -\cot \frac{x}{2}$:

$$f'(x) = \frac{1}{2} \csc^2 \frac{x}{2}$$

This matches our integrand exactly.

$$\int \frac{1 - \sin x}{1 - \cos x} e^x dx = -e^x \cot \frac{x}{2} + C$$

Additional Integrals

Part (a): $\int e^{ax} \cos(bx + c) dx$

Formula:

$$\int e^{ax} \cos(bx + c) dx = \frac{e^{ax} [a \cos(bx + c) + b \sin(bx + c)]}{a^2 + b^2} + C$$

$$\int e^{ax} \cos(bx + c) dx = \frac{e^{ax} [a \cos(bx + c) + b \sin(bx + c)]}{a^2 + b^2} + C$$

Part (b): $\int_0^{\pi/2} \sin^2 \frac{x}{2} \cos^2 \frac{x}{2} dx$

Step 1: Use identity: $\sin^2 \frac{x}{2} \cos^2 \frac{x}{2} = \frac{1}{4} \sin^2 x$

Step 2:

$$\frac{1}{4} \int_0^{\pi/2} \sin^2 x dx = \frac{1}{4} \int_0^{\pi/2} \frac{1 - \cos 2x}{2} dx = \frac{1}{8} \left[x - \frac{\sin 2x}{2} \right]_0^{\pi/2} = \frac{1}{8} \cdot \frac{\pi}{2}$$

$$= \frac{\pi}{16}$$

Part (c): $\int \frac{\ln x}{\sqrt{x}} dx$

Step 1: Integration by parts: $u = \ln x, dv = x^{-1/2} dx \implies v = 2\sqrt{x}$

$$\begin{aligned} \int \frac{\ln x}{\sqrt{x}} dx &= 2\sqrt{x} \ln x - \int 2\sqrt{x} \cdot \frac{1}{x} dx = 2\sqrt{x} \ln x - 2 \int x^{-1/2} dx \\ &= 2\sqrt{x} \ln x - 4\sqrt{x} + C \end{aligned}$$

$$\int \frac{\ln x}{\sqrt{x}} dx = 2\sqrt{x}(\ln x - 2) + C$$

Part (d): $\int x^2 \sec^2 x^2 dx$

Step 1: Integration by parts: $u = x^2$, $dv = \sec^2 x^2 dx$.

Step 2: Let $t = x^2 \implies dt = 2x dx$:

$$\int x^2 \sec^2(x^2) dx = \frac{1}{2} \int t \sec^2 t dt$$

Step 3: By parts: $u = t$, $dv = \sec^2 t dt \implies v = \tan t$:

$$= \frac{1}{2} \left[t \tan t - \int \tan t dt \right] = \frac{1}{2} [x^2 \tan(x^2) + \ln |\cos(x^2)|] + C$$

$$\int x^2 \sec^2(x^2) dx = \frac{x^2 \tan(x^2)}{2} + \frac{\ln |\cos(x^2)|}{2} + C$$

Part (e): $\int_{\sqrt{e}}^e \frac{\ln x}{x^2} dx$

Step 1: By parts: $u = \ln x$, $dv = x^{-2} dx \implies v = -\frac{1}{x}$

$$\int \frac{\ln x}{x^2} dx = -\frac{\ln x}{x} + \int \frac{1}{x^2} dx = -\frac{\ln x}{x} - \frac{1}{x} + C = -\frac{\ln x + 1}{x} + C$$

Step 2: Evaluate:

$$\left[-\frac{\ln x + 1}{x} \right]_{\sqrt{e}}^e = -\frac{1+1}{e} - \left(-\frac{\frac{1}{2}+1}{\sqrt{e}} \right) = -\frac{2}{e} + \frac{3}{2\sqrt{e}}$$

$$\int_{\sqrt{e}}^e \frac{\ln x}{x^2} dx = \frac{3}{2\sqrt{e}} - \frac{2}{e}$$

Evaluate $\int_{-1}^5 |x - 2| dx$

Step 1: Split at $x = 2$:

$$\int_{-1}^2 (2 - x) dx + \int_2^5 (x - 2) dx$$

Step 2:

$$\begin{aligned} & \left[2x - \frac{x^2}{2} \right]_{-1}^2 + \left[\frac{x^2}{2} - 2x \right]_2^5 \\ &= \left[(4 - 2) - \left(-2 - \frac{1}{2} \right) \right] + \left[\left(\frac{25}{2} - 10 \right) - (2 - 4) \right] \end{aligned}$$

$$= \left[2 + \frac{5}{2} \right] + \left[\frac{5}{2} + 2 \right] = \frac{9}{2} + \frac{9}{2}$$

$$\int_{-1}^5 |x - 2| dx = 9$$

$$\int \frac{4e^x + 6e^{-x}}{9e^x + 4e^{-x}} dx$$

Step 1: Write numerator as $\lambda \cdot (\text{denominator}) + \mu \cdot \frac{d}{dx}(\text{denominator})$:

$$\frac{d}{dx}(9e^x + 4e^{-x}) = 9e^x - 4e^{-x}$$

$$4e^x + 6e^{-x} = \lambda(9e^x + 4e^{-x}) + \mu(9e^x - 4e^{-x})$$

Step 2: Compare coefficients:

$$e^x : \quad 4 = 9\lambda + 9\mu \quad e^{-x} : \quad 6 = 4\lambda - 4\mu$$

$$\lambda + \mu = \frac{4}{9}, \quad \lambda - \mu = \frac{3}{2}$$

$$\lambda = \frac{35}{36}, \quad \mu = -\frac{11}{36}$$

Step 3: Integrate:

$$\int \frac{4e^x + 6e^{-x}}{9e^x + 4e^{-x}} dx = \frac{35}{36}x - \frac{11}{36} \ln|9e^x + 4e^{-x}| + C$$

$$\int \frac{4e^x + 6e^{-x}}{9e^x + 4e^{-x}} dx = \frac{35}{36}x - \frac{11}{36} \ln|9e^x + 4e^{-x}| + C$$

ANALYTIC GEOMETRY

GEO Q.1(a) — Line through (3, 4, 5) intersecting z -axis at right angles

Step 1: A line meeting the z -axis perpendicularly has direction $\mathbf{d} = (a, b, 0)$ (no z -component).

Step 2: The line meets the z -axis at point $(0, 0, k)$.

Direction from $P(3, 4, 5)$ to $(0, 0, k)$: $(-3, -4, k - 5)$.

For $\perp$ to z -axis: $k - 5 = 0 \implies k = 5$.

Step 3: Direction: $(-3, -4, 0)$.

Symmetric form:

$$\frac{x - 3}{-3} = \frac{y - 4}{-4}, \quad z = 5$$

GEO Q.1(b) — Plane through $(2, -3, 2)$ and $(1, 2, -1)$, perpendicular to $3x - 2y + z = 7$

Step 1: Normal of given plane: $\mathbf{n}_1 = (3, -2, 1)$.

Step 2: Direction vector in required plane:

$$\mathbf{v} = (1 - 2, 2 - (-3), -1 - 2) = (-1, 5, -3)$$

Step 3: Normal of required plane: $\mathbf{n} = \mathbf{n}_1 \times \mathbf{v}$:

$$\begin{aligned} \mathbf{n} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -2 & 1 \\ -1 & 5 & -3 \end{vmatrix} = \mathbf{i}[(-2)(-3) - (1)(5)] - \mathbf{j}[(3)(-3) - (1)(-1)] + \mathbf{k}[(3)(5) - (-2)(-1)] \\ &= \mathbf{i}(1) - \mathbf{j}(-8) + \mathbf{k}(13) = (1, 8, 13) \end{aligned}$$

Step 4: Plane through $(2, -3, 2)$:

$$1(x - 2) + 8(y + 3) + 13(z - 2) = 0$$

$$x + 8y + 13z - 4 = 0$$

GEO Q.1(c) — Perpendicular bisector plane of segment joining $(3, -2, 5)$ and $(5, 4, -3)$

Step 1: Midpoint:

$$M = \left(\frac{3+5}{2}, \frac{-2+4}{2}, \frac{5+(-3)}{2} \right) = (4, 1, 1)$$

Step 2: Normal (direction of segment):

$$\mathbf{n} = (5 - 3, 4 - (-2), -3 - 5) = (2, 6, -8) \sim (1, 3, -4)$$

Step 3: Plane through $M(4, 1, 1)$:

$$1(x - 4) + 3(y - 1) + (-4)(z - 1) = 0$$

$$x + 3y - 4z - 3 = 0$$

GEO Q.2(a) — Line through $(2, -3, 2)$ perpendicular to two given lines

Step 1: Rewrite lines in standard form:

$$L_1 : 5x = y + 3 = 2 - z \implies \frac{x}{1} = \frac{y+3}{5} = \frac{z-2}{-5} \quad \mathbf{d}_1 = (1, 5, -5)$$

$$L_2 : x + 3 = 1 - 2y = 1 + z \implies \frac{x+3}{2} = \frac{y-\frac{1}{2}}{-1} = \frac{z-1}{2} \quad \mathbf{d}_2 = (2, -1, 2)$$

Step 2: Required direction: $\mathbf{d} = \mathbf{d}_1 \times \mathbf{d}_2$:

$$\mathbf{d} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 5 & -5 \\ 2 & -1 & 2 \end{vmatrix} = \mathbf{i}(10 - 5) - \mathbf{j}(2 + 10) + \mathbf{k}(-1 - 10) = (5, -12, -11)$$

$$\frac{x-2}{5} = \frac{y+3}{-12} = \frac{z-2}{-11}$$

GEO Q.2(b) — Plane containing two lines

Lines:

$$\frac{x+2}{2} = \frac{3-y}{3} = \frac{z}{2}, \quad \frac{x+2}{3} = \frac{y-3}{2} = \frac{z}{1}$$

Step 1: Common point at $t = 0$: $(-2, 3, 0)$.

Step 2: Directions: $\mathbf{d}_1 = (2, -3, 2)$, $\mathbf{d}_2 = (3, 2, 1)$.

Step 3: Normal:

$$\mathbf{n} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -3 & 2 \\ 3 & 2 & 1 \end{vmatrix} = \mathbf{i}(-3 - 4) - \mathbf{j}(2 - 6) + \mathbf{k}(4 + 9) = (-7, 4, 13)$$

Step 4: Plane through $(-2, 3, 0)$:

$$-7(x + 2) + 4(y - 3) + 13z = 0$$

$$-7x + 4y + 13z - 26 = 0$$

GEO Q.2(c) — Show lines intersect; find common point

$$L_1 : \frac{x+1}{2} = \frac{y-2}{2} = \frac{z-5}{-3} = t \implies x = 2t - 1, y = 2t + 2, z = -3t + 5$$

$$L_2 : \frac{x-3}{-1} = \frac{y-2}{-3} = \frac{z}{2} = s \implies x = -s + 3, y = -3s + 2, z = 2s$$

Step 1: Equate x and y :

$$2t - 1 = -s + 3 \implies 2t + s = 4 \quad \dots (i)$$

$$2t + 2 = -3s + 2 \implies 2t + 3s = 0 \quad \dots (ii)$$

Step 2: Subtract (i) from (ii):

$$2s = -4 \implies s = -2, \quad t = 3$$

Step 3: Check z : $L_1 : -3(3) + 5 = -4$; $L_2 : 2(-2) = -4$ ✓

Step 4: Common point: $x = 2(3) - 1 = 5$, $y = 2(3) + 2 = 8$, $z = -4$

Lines **do intersect**. Common point:

$$(5, 8, -4)$$

Additional Plane Problems

Problem 30. Plane through $P_1(-2, 1, 4)$ and $P_2(1, 0, 3)$, perpendicular to $4x - y + 3z = 2$:

$$\mathbf{n}_1 = (4, -1, 3), \quad \mathbf{v} = P_2 - P_1 = (3, -1, -1), \quad \mathbf{n} = \mathbf{n}_1 \times \mathbf{v} = (4, 13, -1)$$

$$4(x + 2) + 13(y - 1) - (z - 4) = 0 \implies \boxed{4x + 13y - z + 3 = 0}$$

Problem 31. Plane through $(-1, 2, -5)$, perpendicular to $2x - y + z = 1$ and $x + y - 2z = 3$:

$$\mathbf{n}_1 = (2, -1, 1), \quad \mathbf{n}_2 = (1, 1, -2), \quad \mathbf{n} = \mathbf{n}_1 \times \mathbf{n}_2 = (1, 5, 3)$$

$$(x + 1) + 5(y - 2) + 3(z + 5) = 0 \implies \boxed{x + 5y + 3z + 6 = 0}$$

Problem 32. Plane containing $(2, 0, 3)$ and line $x = -1 + t$, $y = t$, $z = -4 + 2t$:

Point on line: $Q(-1, 0, -4)$; direction $\mathbf{d} = (1, 1, 2)$; $\mathbf{v} = (3, 0, 7)$; $\mathbf{n} = \mathbf{d} \times \mathbf{v} = (7, -1, -3)$

$$7(x - 2) - (y) - 3(z - 3) = 0 \implies \boxed{7x - y - 3z - 5 = 0}$$

Problem 33. Plane equidistant from $(2, -1, 1)$ and $(3, 1, 5)$:

Midpoint $M = (\frac{5}{2}, 0, 3)$; normal $(1, 2, 4)$

$$\left(x - \frac{5}{2}\right) + 2y + 4(z - 3) = 0 \implies \boxed{2x + 4y + 8z - 29 = 0}$$

Line-Plane Intersections (Problems 25–26)

25. $x = -2, y = 4 + 2t, z = -3 + t$

xy -plane ($z = 0$): $t = 3 \implies (-2, 10, 0)$

xz -plane ($y = 0$): $t = -2 \implies (-2, 0, -5)$

yz -plane ($x = 0$): $x = -2 \neq 0 \implies$ no intersection

26. $x = -1 + 2t, y = 3 + t, z = 4 - t$

xy -plane ($z = 0$): $t = 4 \implies (7, 7, 0)$

xz -plane ($y = 0$): $t = -3 \implies (-7, 0, 7)$

yz -plane ($x = 0$): $t = \frac{1}{2} \implies (0, \frac{7}{2}, \frac{7}{2})$

Problems 22–24

22. Line through origin parallel to $x = t, y = -1 + t, z = 2$:

Direction $\mathbf{d} = (1, 1, 0)$.

$x = t, y = t, z = 0$

23. Line $x = 1 + 3t, y = 2 - t$:

(a) x -axis ($y = 0$): $t = 2 \implies (7, 0)$

(b) y -axis ($x = 0$): $t = -\frac{1}{3} \implies (0, \frac{7}{3})$

(c) $y = x^2$: $(2 - t) = (1 + 3t)^2 \implies 9t^2 + 7t - 1 = 0 \implies t = \frac{-7 \pm \sqrt{85}}{18}$

24. Line $\langle x, y \rangle = \langle 4t, 3t \rangle$ meets circle $x^2 + y^2 = 25$:

$$16t^2 + 9t^2 = 25 \implies 25t^2 = 25 \implies t = \pm 1$$

Points: $(4, 3)$ and $(-4, -3)$

Solved by Hadi Ali – Best of Luck